Freeze-Tag in Any Dimension: Linear Bounds and Geometric Schedules

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Abstract

The freeze-tag problem asks how quickly one active robot can wake a swarm: reaching a sleeping robot activates it immediately, and both robots can then help. We assume unit speed and known sleeper positions within distance at most one of the source. For every norm-induced distance in d dimensions, we prove that the worst-case optimal completion time lies between $d/(e \ln 2)$ and $6d + 2$, establishing linear growth in dimension with no dependence on swarm size. For Euclidean (ℓ_2) distance, we improve the upper bound to $(2/\ln 2)d + O(\sqrt{d \log d})$. We also sharpen linear lower bounds for Euclidean and Manhattan (ℓ_1) distances. Our main approach recruits a concentrated group of sleepers first, then uses them to activate the rest in parallel. It also yields polynomial-time approximation algorithms using only pairwise distances, with weaker constants. More generally, we characterize the metric spaces admitting a uniform completion-time bound proportional to the initial radius: these are exactly the spaces of finite doubling dimension, where every ball can be covered by a fixed number of balls of half its radius. Finally, we improve upper bounds from 4.31 to below 4.16 in the Euclidean plane, from 12 to $197/21 < 9.381$ in three-dimensional Manhattan distance, and from 12.76 to below 8.68 in three-dimensional Euclidean distance. The general-dimensional and three-dimensional Manhattan bounds are analytic; the two low-dimensional Euclidean upper bounds use rigorous computer-assisted proofs.

CCS Concepts

• **Theory of computation** → **Computational geometry**; • **Computing methodologies** → *Multi-agent systems.*

Keywords

Freeze-tag; swarm activation; geometric scheduling; crown recursion; doubling dimension; computer-assisted proof

1 Introduction

One robot is initially awake at a source s, and n others sleep at known positions within distance at most one of s. Awake robots move at unit speed. Reaching a sleeper wakes it immediately, after which both move independently. The *freeze-tag problem* minimizes the time until all robots are awake [1].

We study $\mathbb{R}^d$ with any norm. For such a space X, the *wake-up ratio* γ_X is the supremum of optimal completion times over finite sleeper sets in its unit ball, starting at the center [4]. Scaling gives completion within $\gamma_X H$ for sleepers within distance H, regardless of their number. Write $\gamma_{d,p}$ for the ℓ_p ratio; ℓ_2 and ℓ_1 give Euclidean and Manhattan distance.

Table 1: Bounds on the wake-up ratio.

Space	Lower bound	Previous upper	New upper
$(\mathbb{R}^2, \ell_2)$	$1 + 2\sqrt{2}$	4.31	< 4.16
$(\mathbb{R}^3, \ell_1)$	$17/3$	12	< 9.381
$(\mathbb{R}^3, \ell_2)$	$1 + 6\sqrt{(4 - \sqrt{2})/7} > 4.64$	12.76	< 8.68

Contributions. For every d-dimensional normed space X, we prove $d/(e \ln 2) \le \gamma_X \le 6d+2$, establishing linear growth uniformly over all norms. The Euclidean upper bound improves to $(2/\ln 2)d + O(\sqrt{d \log d})$, and we sharpen the Euclidean and Manhattan linear lower bounds. Our method wakes a group in a smaller ball, then uses it to wake the rest in parallel. Volume averaging [17] finds the group; directional averaging improves the Euclidean bound.

A metric space admits a uniform completion-time bound proportional to the initial radius exactly when it has finite doubling dimension: every ball is covered by a fixed number of half-radius balls. Our distance-only algorithms use $O(n^3)$ arithmetic and comparison operations and give approximation factors linear in the normed-space or doubling dimension, with larger constants than the existence bounds (Section 3.4).

Table 1 compares our low-dimensional bounds with [10]. Planar schedules combine annular sectors with chords. In three dimensions, we improve recruitment and use rectangle sweeps for Manhattan distance and downward half-space sweeps for Euclidean distance. Explicit trees handle small populations.

Previous work. Arkin et al. [1] gave population-independent bounds and a polynomial-time approximation scheme for fixed-dimensional ℓ_p spaces. Our focus is instead on explicit radius-based bounds and their dimension dependence. Bonichon et al. [7] obtained the planar bound 4.6211. Alipour et al. [2] studied geometric freeze-tag at AAMAS; their later papers [3, 10] established the previous upper bounds in Table 1. Our low-dimensional schedules build on their recruitment and sweep methods.

Other work includes approximation algorithms [8, 12, 13], heuristics [11], and NP-hardness in three-dimensional Manhattan distance [14]. Online variants reveal robots over time [15, 16]; distributed variants require discovering unknown positions [5].

Proofs and reproducibility. The dimension and three-dimensional Manhattan bounds are analytic; the two low-dimensional Euclidean upper bounds combine geometric schedules with exact-arithmetic verification. The supplement gives detailed proofs and trees in Sections B–F, and verification code, inputs, and commands in Section G.

2 Model and proof tools

We use centralized planning in a metric space (M, d_M), without obstacles or collisions. A move from x to y costs time $d_M(x, y)$; continuous paths are unnecessary. The *makespan* is the last activation time; $\mathrm{OPT}_M(s, S)$ is its minimum for source s and finite sleeper set S, with $\mathrm{OPT}_M(s, \varnothing) = 0$. Write $B_M(s, r) = \{x : d_M(s, x) \le r\}$ and $\operatorname{diam} E = \sup_{x,y\in E} d_M(x, y)$.

For a normed space X with unit ball B_X, let $\gamma_X(n)$ be the supremum of $\mathrm{OPT}_X(0, S)$ over n-point sets $S \subseteq B_X$. Allowing all finite populations gives $\gamma_X = \sup_{n\ge1} \gamma_X(n)$. For a metric space, let $\Gamma(M)$ be the supremum of $\mathrm{OPT}_M(s, S)/r$ over $s \in M$, $r > 0$, and finite $S \subseteq B_M(s, r)$. Translation and scaling give $\Gamma(X) = \gamma_X$.

Write $\|x\|_p = (\sum_i |x_i|^p)^{1/p}$ for $1 \le p < \infty$ and $\|x\|_\infty = \max_i |x_i|$; the fixed-population ratio is $\gamma_{d,p}(n)$. Let e_i be the ith coordinate unit vector and $|x-y|$ the Euclidean distance. Angles are in radians; $\log_2$ and ln denote binary and natural logarithms. Resolve ties and boundaries consistently. Incidental activations cannot hurt: a robot awakened early may wait for its assigned task.

An *awakening tree* has the source as root and one vertex per sleeper. The root has at most one child and other vertices at most two, corresponding to the robots available after activation. Edges are weighted by distance. The maximum root-path weight is the *weighted depth*; the maximum edge count is the *edge-depth*. Write $x \to y, z$ for two children.

Lemma 2.1. *The optimal makespan is the minimum weighted depth of an awakening tree, and every such tree has edge-depth at least $\lceil \log_2(n+1) \rceil$. If active robots and remaining sleepers lie in a set of diameter D, one round of length at most D doubles the active population or finishes the instance.*

Proof. Record each robot's next activation [1]; initially only one robot is available. Edge lengths do not exceed elapsed times, so root-path weights are bounded by the makespan. Conversely, assigning one robot per child realizes the weighted-depth bound by induction. Edge-depth L permits at most $1 + 2 + \cdots + 2^{L-1} = 2^L - 1$ sleepers. Parallel trips to distinct sleepers either give every active robot a helper or finish the instance. □

For small populations, a *block* is a binary tree on selected sleepers, with deadlines relative to its root's external activation. This activation supplies two robots. Each unused child slot provides an *outgoing job*: a robot available by u can wake a target within distance D by $u + D$. A *template* attaches disjoint blocks at these slots and adds path deadlines; its *capacity* counts sleepers. Vertices in blocks of size at least two are *mandatory*; single-vertex blocks are *ordinary*. A *cluster* is a sleeper set with a specified diameter bound.

Lemma 2.2. *A k-vertex block has $k + 1$ unused slots, available at their parent deadlines. Let i range over the required nonordinary blocks. Suppose block i, of size k_i, can be selected whenever at least h_i sleepers remain, and set $h = \max_i(h_i + \sum_{j<i} k_j)$, with $h = 0$ when there are none. A template of capacity N with q mandatory vertices handles every $\max\{h, q\} \le n \le N$, provided no ordinary vertex has a mandatory descendant.*

Proof. Of $2k$ child slots, $k-1$ are internal, leaving $k+1$ robots for outgoing jobs. Select required blocks first. Before block i, $\sum_{j<i} k_j$ sleepers are reserved, so $n \ge h$ guarantees disjoint selection. The descendant condition supplies an ordinary leaf whenever ordinary vertices remain. Prune these leaves until the template has $n \ge q$ vertices, then assign unreserved sleepers to its ordinary vertices. Pruning preserves mandatory vertices and deadlines; distinct slots prevent using a robot twice. □

Select blocks in decreasing order of $h_i - k_i$. For adjacent blocks in this order, $\max\{h_i, k_i + h_j\} \le k_j + h_i$, no larger than the reverse-order requirement. Exchanging inverted pairs therefore minimizes h.

3 Bounds in arbitrary dimension

3.1 Recruiting a dense subset

The (λ, α) *density property*, for $0 < \lambda < 1$ and $0 < \alpha \le 1$, means that every finite sleeper set in a radius-r ball has at least an α fraction in some radius-λr ball, uniformly over centers and scales. The smaller ball may extend outside the original.

Theorem 3.1 (Density recruitment). *Suppose this property holds, and set $L = \lceil \log_2(1/\alpha) \rceil$ and $K = (2 + 2L)/(1 - \lambda)$. If the source and sleepers lie in a radius-r ball, completion within Kr is possible with all robots left at input positions. Starting at the center saves r, so*

$$\Gamma(M) \le \frac{2 + 2L}{1 - \lambda} - 1. \tag{1}$$

A cover by N smaller balls supplies the property with $\alpha = 1/N$.

Proof. Induct on n for the arbitrary-source bound; $n = 0$ is immediate. Choose $m \ge \lceil \alpha n \rceil$ sleepers in a radius-λr ball and reach one within $2r$. Leave one robot idle while the other wakes the remaining $m - 1 < n$ selected sleepers. This source and its sleepers lie in the smaller ball, so induction bounds the additional time by $K\lambda r$.

The $m + 1$ recruited robots are at input positions in the original ball. Since $2^L \ge 1/\alpha$ and $m \ge \alpha n$, we have $(m + 1)2^L \ge n + 1$. Lemma 2.1 finishes in L rounds of length $2r$, for total time $2r + K\lambda r + 2Lr = Kr$. At the outermost call, a central source shortens the first trip to r, giving (1). Assigning sleepers to covering balls and choosing a largest class gives $\alpha = 1/N$. □

Volume averaging supplies the required dense group in every norm.

Theorem 3.2. *For every d-dimensional normed space X, with $d \ge 1$, we have $d/(e \ln 2) \le \gamma_X \le 6d + 2$ and $\gamma_X < 5.992d + 5.202$.*

Proof. Let B be the unit ball and choose Z uniformly in $(1+\lambda)B$. A sleeper $x \in B$ lies in $Z + \lambda B$ exactly when $Z \in x - \lambda B \subseteq (1 + \lambda)B$, with probability $(\lambda/(1 + \lambda))^d$. Averaging gives a ball containing at least this fraction of sleepers [17, Theorem 1.7]. Apply Theorem 3.1 with $L = \lceil d \log_2(1 + 1/\lambda) \rceil$. Taking $\lambda = 1/3$ gives $L = 2d$ and $6d + 2$; taking $\lambda = 71/200$ and using $\lceil x \rceil \le x + 1$ gives $5.992d + 5.202$.

For $0 < \varepsilon < 1$, choose a maximal set $P \subseteq B$ containing 0 with mutual distances exceeding ε. Compactness makes P finite; maximality makes its radius-ε balls cover B. Volume gives $|P| \ge \varepsilon^{-d}$. Placing sleepers at $P \setminus \{0\}$ gives $|P| - 1$ sleeper vertices, so Lemma 2.1 forces $\lceil \log_2 |P| \rceil$ edges on some root path, each longer than ε. Its time

is at least $\varepsilon \log_2 |P| \geq d\varepsilon \log_2(1/\varepsilon)$. Take $\varepsilon = e^{-1}$. Supplementary Section B.1 gives the general λ-bound and constant calculation. □

3.2 Using Euclidean directions

For $d \geq 2$ and $1/\sqrt{2} \leq \lambda < 1$, put $a = \sqrt{1-\lambda^2}$. For a uniformly chosen unit vector U, the spherical cap probability is $q_d(\lambda) = \Pr(U \cdot e_1 \geq a) = \left(\int_0^{\arcsin \lambda} \sin^{d-2} t\, dt\right) / \left(\int_0^{\pi} \sin^{d-2} t\, dt\right)$, where the dot denotes the Euclidean inner product.

Theorem 3.3. *One may use $L = \lceil \log_2(1/q_d(\lambda)) \rceil$ in (1). With $m = d-1$, the explicit choice $L = \lceil m \log_2(1/\lambda) + \log_2(4\sqrt{m}) \rceil$ is valid. Consequently,*

$$\gamma_{d,2} \leq \frac{2}{\ln 2} d + O(\sqrt{d \log d}). \tag{2}$$

Proof. The radius-λ ball centered at aU contains 0, since $a \leq \lambda$, and a unit point v exactly when $U \cdot v \geq a$. Convexity then includes $[0, v]$, so every sleeper has inclusion probability at least $q_d(\lambda)$. Averaging gives the density property.

Write $J_j = \int_0^{\pi} \sin^j t\, dt$. The substitution $s = \sin t$ bounds the numerator below by $\int_0^{\lambda} s^{m-1}\, ds = \lambda^m/m$. Integration by parts gives $J_{m-1} J_m = 2\pi/m$, while Cauchy–Schwarz makes J_m/J_{m-1} nondecreasing from $J_1/J_0 = 2/\pi$. Thus $J_{m-1} \leq \pi/\sqrt{m}$ and $q_d(\lambda) \geq \lambda^m/(\pi\sqrt{m}) > \lambda^m/(4\sqrt{m})$, proving the choice of L.

Put $\varepsilon = 1 - \lambda$. Using $-\ln(1-\varepsilon) = \varepsilon + O(\varepsilon^2)$, the bound in (1) becomes $2m/\ln 2 + O(m\varepsilon + (\log m)/\varepsilon)$. For large m, choosing $\varepsilon = \sqrt{\log_2 m/m}$ balances the errors and proves (2). Supplementary Section B.1 gives finite-dimensional estimates. □

At fixed dimension, take the smaller bound. A universal density guarantee has $\alpha \leq \lambda^d$, making (1) at least $2d/\ln 2 + 2\sqrt{2d/\ln 2} - 1$. This limits the estimate, not all schedules; see Supplementary Section B.3.

Sharper lower bounds. For $d \geq 2$, we also obtain $\gamma_{d,1} > 1 + 0.54658(d-1)$ and $\gamma_{d,2} > 1 + 0.54070(d-1)$. For N unit-sphere sleepers with mutual distances exceeding δ, Lemma 2.1 gives time at least $1 + \delta\lfloor \log_2 N \rfloor$: the source edge costs one, and the others exceed δ. Spherical-cap estimates supply Euclidean sets; a distance calculation for signed simplex points supplies Manhattan sets. Supplementary Section C gives the packing and probability arguments.

3.3 The metric characterization

The *doubling constant* [19] is the least integer C such that every closed ball is covered by C closed balls of half its radius, with centers in the same space. Its logarithm $\Delta = \log_2 C$ is the *doubling dimension*; set $\Delta = \infty$ if no finite C exists.

Theorem 3.4. *$\Gamma(M) < \infty$ if and only if $\Delta < \infty$, and then $\Delta/2 \leq \Gamma(M) \leq 4\lceil \Delta \rceil + 3$.*

Proof. Use Theorem 3.1 with $\lambda = 1/2$ and $\alpha = 1/C$ for the upper bound. Conversely, suppose $B_M(s, r)$ cannot be covered by $k-1$ half-radius balls. Starting with s, repeatedly select a point outside the half-radius balls centered at previous choices. This produces k points at mutual distances greater than $r/2$. Put sleepers at the $k-1$ points other than s. A root path has at least $\lceil \log_2 k \rceil$ edges, so normalized time exceeds $\frac{1}{2}\log_2 k$. Minimality of finite $C \geq 2$ supplies such a ball for $k = C$; infinite C permits arbitrarily large k. For $C = 1$, the lower bound is zero. □

For a d-dimensional norm, volume and a maximal separated set give $2^d \leq C \leq 5^d$. Supplementary Section B.4 treats the cube.

3.4 Constructive schedules

A dense radius-λr ball may be hard to find, but its sleepers lie in a radius-$2\lambda r$ ball centered at any one of them. Scanning sleeper centers therefore finds at least as many sleepers at twice the radius.

Corollary 3.5. *Using a distance oracle, deterministic schedules can be constructed in $O(n^3)$ arithmetic and comparison operations with $O(n^2)$ stored pair distances. Their completion times are at most $(8.4d + 1.8)H$ in a d-dimensional normed space and $(4\lceil 2\Delta \rceil + 3)H$ in a metric space of doubling dimension Δ, where $H = \max_{x \in S} d_M(s, x)$. These constants are also approximation ratios.*

Proof. Use $\mu = 2/7$ for norms and $\mu = 1/2$ for doubling metrics. Find the sleeper-centered radius-μr ball containing the most sleepers. Reach its center, reserve one robot, recursively wake the group with the other, then finish by doubling rounds. This implements Theorem 3.1. For norms, volume averaging with $\lambda = 1/7$ gives density 8^{-d} at scanned radius $2r/7$, yielding $(2{+}6d)/(1{-}2/7){-}1 = 8.4d{+}1.8$. Two half-radius covers in a doubling metric give density C^{-2} at radius $r/4$, hence scanned radius $r/2$. The same theorem gives $4\lceil 2\Delta \rceil + 3$ without knowing C.

Each call has one recursive child with fewer sleepers. Scans take $O(k^2)$ comparisons, hence $O(n^3)$ overall with stored pair distances. Every activation path costs at least its endpoint distance from the source, so OPT $\geq H$. □

Supplementary Section B.2 gives pseudocode and the randomized directional variant.

4 Planar Euclidean activation below 4.16

Theorem 4.1. *With one active robot at the center of the unit disk, $1 + 2\sqrt{2} \leq \gamma_{2,2} < 4.16$.*

An inscribed square forces a root path with edges of lengths at least $1, \sqrt{2}, \sqrt{2}$. The conjectured upper bound $1 + 2\sqrt{2}$ holds for $n \leq 7$, $n \geq 281$, or convex-position sleepers [7, Theorems 1–3]. Our certificate covers all $n \geq 4$.

4.1 Schedules for an annular sector

A *crown* has radii $[b, 1]$ and angles $[0, \theta]$, with $0 \leq b \leq 1$ and $0 \leq \theta \leq 2\pi$. Its width is $w = 1 - b$. The endpoints of its angle-zero radial side are the *inner* and *outer corners*; they need not contain sleepers. Reflection reverses direction.

Put $\varphi = (1 + \sqrt{5})/2$. The classical construction takes at most $\theta + \varphi w$ from two robots together on the initial radial side, or $\theta + \varphi^2 w$ from one [7]. It recruits a first sleeper in each radial subcrown and recurses; angular travel telescopes as widths shrink. Supplementary Lemma D.1 proves the induction. With outer radius H and absolute width $\widehat{w}$, $\theta + Kw$ scales to $H\theta + K\widehat{w}$.

Let A_k, B_k be width coefficients for two robots at the outer and inner corners. One-robot transfers and two-robot splits improve

each other. Suppressing the argument θ, define

$$A_0 = \frac{\varphi^4}{\varphi^3+\theta}, \qquad B_0 = 1+\frac{\varphi^2}{\varphi^3+\theta}, \tag{3}$$

$$A_{k+1} = \frac{(1+B_k)^2}{1+2B_k+\theta}, \qquad B_{k+1} = \frac{(1+A_k)^2+\theta}{1+2A_k+\theta}. \tag{4}$$

Here k counts refinements, not activation rounds. The coefficient A_0 is from [7, Lemma 5]; its one-robot inner-corner consequence is extended to all angles in [10, Lemma 3].

Theorem 4.2 (Corner bounds). *For every $k \ge 0$, two robots at a crown's inner or outer corner finish within $\theta + B_k(\theta)w$ or $\theta + A_k(\theta)w$, respectively. One robot at the inner or outer corner finishes within $\theta + (1+A_k(\theta))w$ or $\theta + (1+B_k(\theta))w$, respectively.*

Proof. The recurrence satisfies

$$0 < A_k \le B_k \le \varphi, \quad B_k \ge 1, \quad B_k - A_k < 1,$$
$$-2/3 \le A_k' \le 0, \qquad -1/2 \le B_k' \le 0. \tag{5}$$

Supplementary Section D.1 proves these inequalities independently of the schedules. Let a nonempty crown's angularly first sleeper have angle ψ and radius ρ, and put $u = \theta - \psi$. Then $A_k(u) - A_k(\theta) \le 2\psi/3$ and $B_k(u) - B_k(\theta) \le \psi/2$. From the inner corner, follow the inner arc to ψ, wake the sleeper, and take both robots to radius one. The bound is $(1-w)\psi + w + u + A_k(u)w \le \theta + (1+A_k(\theta))w$, since $A_k' \ge -1$.

From the outer corner, move radially to ρ and then along its circle to the sleeper. Put $d_r = 1-\rho$. If $d_r \ge w/2$, send the pair inward, giving $\theta + w - d_r\psi + B_k(u)w \le \theta + (1+B_k(\theta))w$. Otherwise, send it outward, giving $\theta + d_r(2-\psi) + A_k(u)w$. For $\psi \le 2$, use $d_r \le w/2$ and $A_k(u) \le B_k(\theta) + \psi/2$; for $\psi > 2$, use $d_r(2-\psi) \le 0$ and $A_k(u) \le \varphi < 2 \le 1 + B_k(\theta)$. Both cases give the claimed outer-corner bound.

Conversely, split off an outer strip of width zw and apply scaled one-robot bounds. For an outer pair, put $P = 1 + B_k(\theta)$. One robot handles the strip; the other moves inward by zw. Their width coefficients Pz and $z(1-\theta) + P(1-z)$ agree at $z = P/(2P+\theta-1)$. For an inner pair, put $P = 1 + A_k(\theta)$; the coefficients $P(1-z) - z\theta$ and $1 + (P-1)z$ agree at $z = (P-1)/(2P+\theta-1)$. Both cuts lie in $[0,1]$ and give (4). Starting from $P = \varphi^2$ instead gives (3). Induction on k proves every finite refinement; the elementary recursion terminates because each nonempty stage activates a sleeper. □

Fix eight refinements and write $C_{\mathrm{i}}(b,\theta) = \theta + (1+A_8(\theta))(1-b)$ and $C_{\mathrm{o}}(b,\theta) = \theta + (1+B_8(\theta))(1-b)$ for the one-robot bounds. Both decrease with b and increase with θ. Write $\mathcal{D}(b,c;\omega) = \sqrt{b^2+c^2-2bc\cos\omega}$ for the distance between radii b, c at angular separation ω.

4.2 Using a chord after recruitment

For a crown of angle u, an outer-corner pair splits at β: one robot clears $[0,\beta]$ from the outer corner, while the other crosses a chord to the inner corner at β and clears $[\beta,u]$. This costs at most $\max\{C_{\mathrm{o}}(b,\beta), \mathcal{D}(1,b;\beta) + C_{\mathrm{i}}(b,u-\beta)\}$. With one robot, first recruit a helper.

Lemma 4.3 (Chordal continuation). *For $0 \le b \le 1$, $0 \le \theta \le 2\pi$, and $\beta, \psi_0 \ge 0$ with $\beta + \psi_0 \le \theta$, one robot at the inner corner has completion bound*

$$\begin{aligned}\widehat{C}(b,\theta;\beta,\psi_0) = \max\{&b\psi_0 + 1 - b + C_{\mathrm{o}}(b,\beta),\\ &1 - b + \mathcal{D}(1,b;\beta)\\ &\quad + C_{\mathrm{i}}(b,\theta-\beta),\\ &b\psi_0 + C_{\mathrm{i}}(b,\theta-\psi_0)\}.\end{aligned} \tag{6}$$

For fixed angles, this expression is convex in b.

Proof. For a nonempty crown, let the angularly first sleeper have angle ψ and radius ρ. Follow the inner arc to ψ, wake it, and continue radially with both robots to the outer circle, costing $b\psi + (\rho - b) + (1-\rho) = b\psi + 1 - b$.

If $\psi \le \psi_0$, the split at $\psi + \beta$ is legal because $\beta + \psi_0 \le \theta$. The first branch is bounded by the first entry of (6); the second costs at most $b\psi + 1 - b + \mathcal{D}(1,b;\beta) + C_{\mathrm{i}}(b,\theta-\psi-\beta)$. By (5), $\partial_u C_{\mathrm{i}}(b,u) \ge (1+2b)/3 \ge b$, so this cost is nonincreasing in ψ. Its value at zero gives the second entry.

If $\psi \ge \psi_0$, use the outer-pair bound $\theta - \psi + A_8(\theta-\psi)(1-b)$. Including recruitment gives $b\psi + C_{\mathrm{i}}(b,\theta-\psi)$, again nonincreasing in ψ, so ψ_0 gives the third entry. Every entry is affine in b except for a chord, which is a norm of an affine function. Their maximum is convex. Empty crowns need no motion. □

4.3 Recruiting the initial teams

Use the recruitment patterns of [10]. Since $n \le 3$ finishes by time 3, assume $n \ge 4$. Assign arbitrary angles at the origin. The three nearest sleepers p_1, p_2, p_3 have radii $a \le r \le R$ and, after rotation and reflection, angles $0, t, v$, with $0 \le t \le \pi$. Let ω be the largest absolute angle excluding p_1, p_2. Then $|v| \le \omega \le \pi$; after these seeds are awake, all remaining sleepers lie in $[R,1] \times [-\omega, \omega]$.

Two crowns from one activation. Wake p_1 and continue both robots along its ray to radius r, arriving at time r. All remaining sleepers have radius at least r. For $\max\{t,\omega\} \le \Theta \le \pi$, assign the positive- and negative-angle crowns to the two robots. The completion bound is either $r + C_{\mathrm{i}}(r,\Theta)$ or $r + \widehat{C}(r,\Theta;\beta,\psi_0)$.

Four crowns from three activations. Wake p_i and send its robots to p_j, p_ℓ. Pair $k \in \{j,\ell\}$ is ready by $\tau_k = |p_i| + |p_i - p_k|$. Fix its corner choice $(h_k, G_k) \in \{(R, C_{\mathrm{i}}), (1, C_{\mathrm{o}})\}$. For polar coordinates (r_k, θ_k), put $F_k(\beta,u) = \tau_k + \mathcal{D}(r_k, h_k; \theta_k - \beta) + G_k(R,u)$: recruitment, corner travel, and an angle-u crown.

Choose $\omega \le \Omega \le \pi$ and $0 \le x \le \Omega$. Assign $[-\Omega, 2x-\Omega]$ and $[2x-\Omega, \Omega]$ to the two pairs. Each starts at its interval's midpoint and sends one robot in each direction, giving $\max\{F_j(x-\Omega, x), F_\ell(x, \Omega - x)\}$. Alternatively, pair j clears $[-x,x]$ and pair ℓ clears the two exterior intervals, giving $\max\{F_j(0,x), F_\ell(-\Omega, \Omega-x), F_\ell(\Omega, \Omega-x)\}$. The full-annulus version uses antipodal rays $\xi, \xi+\pi$ and bound $\max\{F_j(\xi,x), F_\ell(\xi+\pi, \pi-x)\}$. Each pair starts at its own ready time.

Other schedules. The certificate also uses three-crown, far-sleeper, and cleanup schedules. For three crowns, one robot from p_1 recruits p_2 or p_3; the pair clears two angle-x crowns, while the spare clears angle $2\pi - 2x$. A far-sleeper schedule wakes p_2, then p_1 and an angularly farthest eligible q at angle $\pm\omega$, radius $\rho \in [R,1]$. Pair q

chooses its corner by total remaining time. Cleanup assigns two crowns to one pair and includes separate seed trips in the makespan. Supplementary Section D.3 specifies the trees, rays, and branch formulas.

4.4 Bounding every configuration

Place a in a band $jr/16 \le a \le (j+1)r/16$, $0 \le j < 16$. Recruitment bounds are monotone or convex in a, so endpoints suffice. The remaining domain is $0 \le r \le R \le 1$, $0 \le t \le \pi$, $|v| \le \omega \le \pi$. Bands and signs of v give 32 roots. Bisect until each leaf has a schedule valid throughout its box, choosing enclosing angles at upper endpoints.

Lemma 4.4 (Continuous-domain certificate). *For each feasible box B, fix one of these schedules, including its corners and cut angles. Compute an upper bound $U(B)$ from radial endpoints, upper endpoints for enclosing angles, and lower bounds for distance cosines. For a far sleeper's corner choice with crown angle u, also test $\rho_* = (1+R+C_o(R,u)-C_i(R,u))/2$. If finitely many such boxes cover the domain and each $U(B) < 4.16$, then every instance finishes below 4.16 with the same verified margin.*

Proof. For fixed angles, distances are norms of affine radial functions; crown costs are affine. Sums and maxima are convex, so radial rectangle corners suffice, even if infeasible. A lower cosine bound in $[-1,1]$ preserves this norm representation; clip smaller bounds to -1. The triangle inequality makes $a + \mathcal{D}(a,\rho;\zeta)$ nondecreasing and $\mathcal{D}(a,\rho;\zeta) - a$ nonincreasing in a, covering the remaining nearest-radius terms.

The far-sleeper branch is bounded by

$$r + \max_{R\le\rho\le 1}\Big[\mathcal{D}(r,\rho;t+\omega) + \min\{\rho - R + C_i(R,u), \; 1-\rho+C_o(R,u)\}\Big].$$

Since $\cos(t-\omega) \ge \cos(t+\omega)$, the angle $t+\omega$ covers either sign. The affine corner costs meet at $\rho_* \in [R,1]$, since $0 \le B_8 - A_8 < 1$. On either side, the chosen cost plus distance is convex, so test only R, ρ_*, and 1. Substituting the affine $\rho_*(R)$ preserves convexity in the remaining radii. This does not assume convexity of the minimum.

Larger enclosing arcs and smaller cosines increase the bounds. A cosine minimum occurs at an interval endpoint or an included odd multiple of π. These reductions cover the whole box, including all first-sleeper positions in Lemma 4.3. The largest verified enclosure proves the claim; Supplementary Section D.4 supplies the endpoint details. □

The C++ checker reconstructs 32 roots and every bisection: 114 050 nodes and 57 041 leaves, at scale 2^{56}. Its largest bound is $299759586229607739/2^{56} = 4.159999931052\ldots < 4.16$. A separate Python implementation reconstructs the domain and checks every leaf at scale 2^{72}. Supplementary Section D.5 gives the largest-bound leaf, verified margin, rounding rules, and reproduction details.

5 Three-dimensional ℓ_1 activation

Theorem 5.1. *For one source at the center of the unit ℓ_1 ball, $17/3 \le \gamma_{3,1} \le 197/21 < 9.381$. The upper bound has an analytic proof using explicit finite scheduling trees.*

Recruit eight robots for cylinder sweeps; use explicit trees for smaller populations.

5.1 Recruiting eight robots

The unit ℓ_1 ball is covered by 27 balls of radius $3/7$, centered at

$$\begin{aligned} C = {} & \{0\} \cup \{\pm 4e_i/7 : 1 \le i \le 3\} \\ & \cup \{2(\pm e_i \pm e_j)/7 : i < j\} \\ & \cup \{2(\pm e_1 \pm e_2 \pm e_3)/7\}. \end{aligned} \tag{7}$$

By symmetry, take (a,b,c) with $a \ge b \ge c \ge 0$ and $s = a+b+c \le 1$. The origin covers $s \le 3/7$. Otherwise, if $a < 2/7$, use $(2/7,2/7,2/7)$, at distance $6/7 - s < 3/7$. For $b \ge 2/7$, use $(2/7,2/7,0)$; for $a \ge 4/7$, use $(4/7,0,0)$. Both distances are $s - 4/7 \le 3/7$. Otherwise, $2/7 \le a < 4/7$ and $b < 2/7$. Distances to $(4/7,0,0)$ and $(2/7,2/7,2/7)$ are $4/7 + s - 2a$ and $2/7 + 2a - s$, summing to $6/7$, so one is at most $3/7$.

For $n \ge 163 = 27(7-1)+1$, a covering ball contains seven sleepers. Reach one by time 1, then double twice within this diameter-$6/7$ group: 2 robots become 4, then 8, by time $19/7$. All return to the origin by $26/7$, since their input positions lie in the unit ball, even if the covering ball extends outside it.

5.2 An analytic cylinder sweep

The coordinates $u = x+y$, $v = x-y$ satisfy $|\Delta x|+|\Delta y| = \max\{|\Delta u|, |\Delta v|\}$, turning a radius-$\rho$ horizontal diamond into a side-2ρ square in ℓ_∞. Let $F(A,B)$ bound horizontal travel from two robots together anywhere in an $A \times B$ rectangle, with sleepers at relative heights in $[0,h]$, $h \ge 0$. Upward height increments sum to at most h, giving total time $F(A,B) + h$.

Assume $A \ge B$ and reflect the longer coordinate so the start lies in its first half. Cut at $a \in [A/2, A)$ and send one robot to a lowest sleeper in each nonempty child. The near branch costs at most $\max\{a,B\}$ horizontally; the far branch costs at most A. Each arrival supplies a pair with no lower assigned sleepers. It suffices that

$$F(A,B) \ge \max\{\max\{a,B\} + F(a,B),\ A + F(A-a,B)\}.$$

Sort child sides before evaluating F. Each arrival removes a sleeper, so population induction proves the sweep without returns to rectangle centers.

Lemma 5.2. *Put $\kappa = (\sqrt{17}-1)/2$ and $c = 2(\kappa+1)/3$. For $A \ge B$, the function $F_*(A,B) = \max\{c(A+B), \kappa A + 2B\}$ is a valid budget, with $F_*(A,A) = (\kappa+2)A$.*

Proof. For $A > 0$, scale to $A = 1$, put $x = B/A$, $\beta_0 = \kappa/(2\kappa+1)$, and $P(x) = F_*(1,x)$. The defining affine terms agree at $x = 1/2$: $P(x) = c(1+x)$ below and $P(x) = \kappa + 2x$ above. Choose the relative cut $q = a/A$ by

$$q(x) = \begin{cases} 3/5, & 0 \le x \le 1/4, \\ (\kappa+1)/(2\kappa+1), & 1/4 < x \le \beta_0, \\ (3+(\kappa-2)x)/(\kappa+3), & \beta_0 < x \le 3/5, \\ (3-x)/4, & 3/5 < x \le 1. \end{cases}$$

All cuts lie in $[1/2, 1)$. Both child widths exceed x on the first two intervals, only q does on the third, and neither does on the fourth, allowing endpoint equality. With side ordering fixed, branch costs are maxima of affine functions. Since P changes slope only at $1/2$, interval endpoints and that breakpoint suffice. Use $\kappa^2 + \kappa = 4$; see Supplementary Section E.2.

On $3/5 \le x \le 1$, both branches cost $H(x) = 3/2 + (\kappa + 1/2)x$, and $H(x) - P(x) = (\kappa - 3/2)(x - 1) \le 0$. Equality for the third cut at $x = 1/2$ explains κ. Scaling and population induction finish the proof; $A = 0$ needs no horizontal motion. □

A single robot at the bottom center reaches a lowest sleeper at horizontal cost at most ρ; its pair then uses $2(\kappa+2)\rho$. Upward travel gives $T_{\mathrm{cyl}}(\rho, h) \le (4 + \sqrt{17})\rho + h$.

Send two robots to each of $(\pm 1/2, 0, 0)$ and $(0, \pm 1/2, 0)$, costing $1/2$. Their radius-$1/2$ diamonds cover the horizontal projection. At each center, one robot sweeps up and one down, each through height at most 1. For $n \ge 163$, this gives $T \le 26/7 + 1/2 + (4 + \sqrt{17})/2 + 1 = (101 + 7\sqrt{17})/14 < 197/21$, since $3\sqrt{17} < 13$.

5.3 Small populations

Besides (7), use eighteen radius-$1/2$ balls centered at $\{\pm e_i/2\} \cup \{(\pm e_i \pm e_j)/4 : i < j\}$ and six radius-$2/3$ balls at $\pm e_i/3$. Ordered-coordinate case checks prove coverage (Supplementary Section E.1). Thus $27(k-1)+1$, $18(k-1)+1$, or $6(k-1)+1$ sleepers guarantee a k-point cluster of diameter $6/7$, 1, or $4/3$, respectively.

Sort by the first coordinate and apply the Erdős–Szekeres theorem [6] to the other two. The successive size guarantees are $17 \to 5 \to 3$ and $82 \to 10 \to 4$, giving three- and four-point coordinatewise monotone chains. Each coordinate may have its own direction. The chain's ℓ_1 length equals its endpoint distance, at most 2; reversing it makes the first edge at most 1.

In a diameter-D cluster, use a breadth-first tree with root depth zero. A slot whose parent has depth j wakes its next target within $jD + 2$ of the root activation; a triple supplies four delays $D + 2$. A monotone triple supplies $2, 3, 4, 4$: one robot leaves while the other follows the chain. A four-point chain adds delay 4. Lemma 2.2 composes these blocks.

Table 2 lists each template's minimum population $n_{\min}$, capacity N, mandatory count q, and completion bound T. Selection and pruning cover every $n_{\min} \le n \le N$.

Table 2: Small-population templates in ℓ_1^3.

$n_{\min}$	N	q	$21T$	$n_{\min}$	N	q	$21T$
1	31	0	189	122	128	114	197
17	39	3	189	129	134	123	197
37	63	15	196	135	138	130	197
63	79	39	196	139	147	134	197
73	95	63	196	148	154	140	197
96	110	88	196	155	160	149	197
111	121	105	197	161	164	155	197

The last eight templates give $197/21$. Let N_k, C_k, and W_k denote breadth-first blocks of k sleepers with diameter bounds $6/7$, 1, and $4/3$. Let M_k be a monotone chain and O an ordinary vertex. Child lists attach roots to unused slots in increasing delay order; powers mean consecutive disjoint copies, and omitted children leave slots empty. Define

$$
\begin{aligned}
Q &= W_3(O^4), & X_1 &= N_5(Q, Q, W_7^4),\\
X_2 &= C_6(C_6, W_3^6), & X_3 &= M_4(C_6, W_3, O^3),\\
X_4 &= C_3(W_3^3, W_2), & X_5 &= M_3(O(O, O), W_2, O, O),\\
T_{13} &= M_4(X_1, X_2, X_3, X_4, X_5).
\end{aligned}
$$

This template has 121 sleepers, 105 mandatory, and threshold $h = 111$ under decreasing threshold-minus-size order. Its longest path costs $1 + 2 + 26/7 + 8/3 = 197/21$: the source trip, the first M_4 job, a late N_5 job, and two W_7 edges. Pruning covers $111 \le n \le 121$.

Supplementary Section E.3 specifies the remaining templates. Together with the sweep for $n \ge 163$, they prove the upper bound in Theorem 5.1.

5.4 The lower bound

Use the six coordinate vertices and $\pm(1/3, 1/3, 1/3)$ from Chun's packing [9, Proposition 5.1]. Source distances are one; sleeper distances are $4/3$ or 2. The distance-$4/3$ pairs form two disjoint three-leaf stars. Eight sleepers force a four-edge root path, but a simple star path has at most two edges. Hence one of the first three sleeper edges has length 2, giving OPT $\ge 1 + 4/3 + 4/3 + 2 = 17/3$.

A matching tree is $0 \to e_1$, $e_1 \to u, -e_1$, $u \to e_2, e_3$, $-e_1 \to -u, -e_2$, and $-u \to -e_3$, where $u = (1/3, 1/3, 1/3)$. Its deepest weighted path has length $17/3$. Supplementary Section E.4 proves the upper bound $17/3$ for every instance with at most nine sleepers by an analytic case split on six diameter-$4/3$ cells.

6 Three-dimensional Euclidean activation

Theorem 6.1. *For the Euclidean unit ball, $\gamma_{3,2} < 8.68$. For $n \le 63$, the analytic bound is $\gamma_{3,2}(n) \le 5 + 2\sqrt{3} < 8.465$.*

After handling small populations, we recruit four robots and bound their downward sweeps using the seed positions and Euclidean travel costs.

6.1 Small populations

Assign unit directions to nearest vertices of a centered regular icosahedron. Each cone is generated by five incident outward unit facet normals $f_1, \dots, f_5$, with $f_i \cdot f_j \ge 1/3$. Nonnegative combinations $x = \sum_i a_i f_i$, $y = \sum_j b_j f_j$ satisfy $x \cdot y \ge \frac{1}{3}(\sum_i a_i)(\sum_j b_j) \ge \frac{1}{3}|x||y|$. Thus unit directions in a cone are at most $2/\sqrt{3}$ apart. Extend cells to the origin, assigning ties consistently. Squared distance is convex in each radius; its corner values on $[0, 1]^2$ are at most $4/3$. The twelve cells therefore have diameter at most $d_* := 2/\sqrt{3} < 5/4$, so $12(k-1)+1$ sleepers guarantee a k-point cluster.

Thirteen templates cover $32 \le n \le 63$ by $35/4$, with cluster edges bounded by $5/4$ and other sleeper edges by 2. For example, a six-point root cluster supplies one early and six later jobs. The early job activates a four-point cluster, whose jobs activate three triples and two ordinary sleepers. Each later job activates a triple with four ordinary children: $6 + (4 + 3 \cdot 3 + 2) + 6(3 + 4) = 63$ sleepers. Supplementary Section F.1 gives the deadlines and selection requirements.

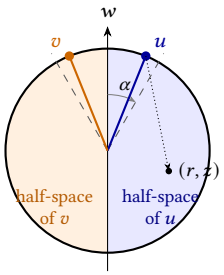


Figure 1: Facing half-spaces in the unit ball.

Replace the cluster-edge bound $5/4$ by d_* in these trees. A path with a cluster edges and b other sleeper edges satisfies $4+5a+8b \le 35$, including the source trip, in units of $1/4$. For $b = 0, 1, 2, 3$, the new sleeper-edge totals are at most $6d_*$, $4d_*+2$, $3d_*+4$, and d_*+6, all at most $3d_*+4$ since $1 \le d_* \le 4/3$. The source trip gives $5+2\sqrt{3}$.

Ordinary doubling handles $n \le 15$ by time 7. For $9 \le n \le 23$, an orthant contains a pair at distance at most $\sqrt{2}$. Wake one by time 1, leave a spare, and send the other robot to its partner. By $1+\sqrt{2}$, three robots can each take at most seven of the $n-2 \le 21$ remaining sleepers; three length-2 rounds suffice. For $17 \le n \le 31$, an orthant triple supplies four robots by $1+\sqrt{2}$, followed by three doubling rounds. These ranges cover $n \le 31$ within $7+\sqrt{2} < 5+2\sqrt{3}$. Supplementary Section F.1 gives the facet calculation and remaining templates.

6.2 Facing half-spaces

Relative to an axis through the origin, let r be horizontal radius and z height. A quadrant spans angle $\pi/2$. Facing half-spaces (Figure 1) avoid first trips to the opposite side.

Lemma 6.2 (Facing assignment). *Let two pairs have nonzero positions with unit directions u, v and $\angle(u,v) \le 2\alpha < \pi$. Put $c = \cos\alpha$ and choose axis $w = (u+v)/|u+v|$. Assign each pair the half-space facing its horizontal projection, split into two quadrants. For a target (r,z) in that half-space, the first trip from a unit-radius seed is at most $H_c(r,z) = \max\{\sqrt{r^2+z^2+1-2cz}, \sqrt{r^2+(z-1)^2}\}$.*

Proof. Put $\beta = \frac{1}{2}\angle(u,v) \le \alpha$. Relative to w, the directions have axial component $\cos\beta$ and opposite horizontal projections. For $u \ne v$, assign the half-spaces $\{x : x \cdot (u-v) \ge 0\}$ and $\{x : x \cdot (u-v) \le 0\}$ to u and v, respectively. Their boundary contains w because $(u+v)\cdot(u-v) = 0$.

The target's horizontal dot product with its assigned seed is nonnegative. For a seed of radius s, squared distance is therefore at most $r^2+z^2+s^2-2sz\cos\beta$. At $s=1$, use $\cos\beta \ge c$ for $z \ge 0$ and $\cos\beta \le 1$ otherwise. If $u = v$, any dividing plane containing w works. □

For fixed x, u with $|u| = 1$ and $A \ge 1$, $As+|su-x|$ is nondecreasing: increasing s by ε adds $A\varepsilon$ while reducing distance by at most ε. The same holds for our distance majorants. Thus maximize recruitment plus first-trip cost together, not distance alone, at the largest radius.

Put $t_0 = 419/1000$, and let n_{in} count sleepers with norm at most t_0; the others lie in the shell $t_0 < |x| \le 1$. For $n \ge 64$, we use certified sphere covers by 31 and 62 caps, with half-angle cosines $c_{31} = 911574/10^6$ and $c_{62} = 954643/10^6$.

If $n_{\text{in}} \ge 3$, wake three inner sleepers to obtain four robots by $3t_0$. Choose an axis through a nonzero pair; assign the other its facing half-space and the axial pair the complement. Horizontal dot products are nonnegative and seed norms at most t_0, so each first trip is bounded by $H_{\text{in}}(r,z) = \sqrt{r^2+(|z|+t_0)^2}$. Total cost is $3t_0 + H_{\text{in}}$.

If $n_{\text{in}} = 2$, wake both inner sleepers and a shell sleeper of radius s. The inner pair has the preceding bound. The shell pair is ready by $2t_0+s$; choose its direction as axis. Radial monotonicity gives $2t_0 + 1 + H_{\text{ax}}$, where $H_{\text{ax}}(r,z) = \sqrt{r^2+(z-1)^2}$.

If $n_{\text{in}} = 1$, two of at least 63 shell sleepers share one of the 62 caps, so their angular separation is at most twice its half-angle. Recruit them after the inner activation and apply Lemma 6.2 with the bisector axis, giving $2t_0 + 1 + H_{c_{62}}$.

If $n_{\text{in}} = 0$, one of the 31 caps contains three sleepers. Wake one of minimum radius a. For a later seed of radius $s \ge a$, monotonicity gives $a + \sqrt{a^2+s^2-2as\cos(2\alpha)} \le s(1+2\sin\alpha)$, where $\alpha = \arccos c_{31}$. Facing assignment and radial monotonicity give $1+2\sqrt{1-c_{31}^2}+H_{c_{31}}$. Pairs start when ready; only the first case may leave inner sleepers.

6.3 Verifying the continuation

Each robot wakes a highest sleeper in its quadrant. The pair splits the quadrant and sends one robot to a highest sleeper in each child, repeating with nonincreasing heights. A state $S = (\theta_S, a, b)$ records sector angle and radii $[a,b] \subseteq [0,1]$. Its policy chooses angular bisection, a radial cut, or a terminal rectangle; $C(S)$ denotes its children.

For $\theta_S \le \pi/2$, radial endpoint maximization gives

$$\operatorname{diam} S = \max\{2b\sin(\theta_S/2), \sqrt{a^2+b^2-2ab\cos\theta_S}\}.$$

Reaching either child's first sleeper costs at most $\operatorname{diam} S$ horizontally, giving $U(S) = \operatorname{diam} S + \max_{T\in C(S)} U(T)$ at nonterminal states. At a terminal state, enclose the sector, up to rotation, in $[a\cos\theta_S, b] \times [0, b\sin\theta_S]$. For sides $A \ge B > 0$, bisect the longer side until $B \le A \le 2B$, charging each diameter. Two further cuts halve both sides, so the remaining geometric tail is $2(\sqrt{A^2+B^2} + \sqrt{A^2/4+B^2})$. Include both balancing charges and this tail in $U(S)$. If $B = 0$, interval bisection costs $2A$. Euclidean travel is at most horizontal travel plus descent. Descents telescope: later radii are at least a, so heights are at least $-\sqrt{1-a^2}$ and total descent at most $z+\sqrt{1-a^2}$.

Write $\mathcal{D}_\theta(r,v,\zeta) = \sqrt{r^2+v^2-2rv\cos\theta+\zeta^2}$. Divide heights into 1024 bands $[\ell_i, u_i] = [-1+i/512, -1+(i+1)/512]$, $i = 0,\dots,1023$, assigning shared boundaries consistently. Intersect each band's ball and, when applicable, shell radial enclosure with $[a,b]$. Let $E_{S,i}$ contain the resulting endpoints; omit empty intervals.

Lemma 6.3 (Height-band reduction). *After an activation, with two robots present, define real functions for every $r \ge 0$, including infeasible radii, by $V^0_{S,i}(r) = \max\{0, U(S)+u_i+\sqrt{1-a^2}\}$. For $h \ge 1$*

further policy steps before the tail estimate, set

$$V^h_{S,i}(r) = \max\Big\{0, \max_{\substack{T\in C(S),\ j\le i\\ v\in E_{T,j}}} \big[\mathcal{D}_{\theta_S}(r, v, u_i - \ell_j) + V^{h-1}_{T,j}(v)\big]\Big\}. \tag{8}$$

Terminal states use V^0 at every depth; take the inner maximum over no eligible child as zero. These functions bound the remaining activation time and are convex in r. Adding recruitment and the first-trip bound, then maximizing over radial and height endpoints, bounds the complete schedule.

Proof. Induct on h for validity and convexity. The depth-zero tail is constant in r. After a split, a highest sleeper in a nonempty child lies in band $j \le i$, with height drop at most $u_i - \ell_j$ and angular separation at most θ_S. Induction bounds travel plus continuation by $\mathcal{D}_{\theta_S}(r, v, u_i - \ell_j) + V^{h-1}_{T,j}(v)$. Convexity in v reduces its maximum to radial endpoints, giving (8). Each resulting term is convex in r, as is their maximum. Empty children require no work.

For recruitment-plus-first-trip bound $\mathcal{R}(r, z)$ and initial quadrant Q, the complete schedule is bounded by

$$\max_{\substack{i,\ r\in E_{Q,i}\\ z\in\{\ell_i,u_i\}}} \{\mathcal{R}(r, z) + V^9_{Q,i}(r)\}.$$

The functions H_c, H_{ax} are norms or maxima of norms of affine functions; $H_{\mathrm{in}}(r, z) = \max_{\pm} \sqrt{r^2 + (z \pm t_0)^2}$ is also convex. Thus the combined bound is convex in (r, z), so rectangle corners suffice. Real majorants remain defined at infeasible corners; directed rounding follows this reduction. □

The checker verifies cap coverage, sector policies, and continuation bounds on the full ball or closed shell. Depth $h = 9$ includes the quadrant split and eight further transitions. Table 3 reports the bounds rounded upward to six decimal places.

Table 3: Certified upper bounds in ℓ^3_2.

Recruitment case	Upper bound
$n_{\mathrm{in}} \ge 3$	8.257073
$n_{\mathrm{in}} = 2$, inner pair	8.245883
$n_{\mathrm{in}} = 2$, axial pair	8.661546
$n_{\mathrm{in}} = 1$	8.678976
$n_{\mathrm{in}} = 0$	8.679546

Combining the maximum certified value $8.679545937580 < 8.68$ with the bound for $n \le 63$ proves Theorem 6.1. Square roots are rounded upward and subtracted cross-terms downward; Supplementary Section F.3 specifies the radial enclosures, denominator-10^{12} numerators, policy inputs, and rounding rules.

An analytic lower bound. Put $\tau = 1/(1 + 2\sqrt{2})$. Place four unit-sphere sleepers at height $\sqrt{\tau}$ and quarter-turn longitudes, and four at $-\sqrt{\tau}$, rotated by $\pi/4$. In this square antiprism [18], neighboring same-ring inner products are τ; the largest cross-ring product is $(1 - \tau)/\sqrt{2} - \tau = \tau$, with all other products smaller. Thus the minimum chord is $d_8 = \sqrt{2 - 2\tau} = 2\sqrt{(4 - \sqrt{2})/7}$. Eight sleepers force a root path with at least four edges, giving $\gamma_{3,2} \ge 1 + 3d_8$ by Lemma 2.1. Supplementary Section F.4 gives a matching tree.

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